Let $u \in N(T)$

$\Rightarrow u \in U : T(u) = 0_V$

$\Rightarrow T(u) = T(0_U)$

$\Rightarrow u = 0$

$\Rightarrow N(T) = \{0\}$

(iii) T is one-one

Let $u(\neq 0) \in N(T)$

$\Rightarrow u \in U : T(u) = 0_V$

$T(0_U) = 0_V$

$\therefore T(u) = T(0_U)$

$\Rightarrow u = 0_U$

$\therefore N(T) = \{0\}$

S.P:- Let $N(T) = \{0\}$

Claim:- T is one-one

Let $u_1, u_2 \in U$ st

$T(u_1) = T(u_2)$

$\Rightarrow T(u_1 - u_2) = 0$

$\Rightarrow u_1 - u_2 \in N(T)$

$\Rightarrow u_1 - u_2 = 0$

$\Rightarrow u_1 = u_2$

$\therefore$ T is one-one.

P Kalika Maths